Missed Codes Review

Lesson Plan. Area of Operation I, II, III, and XII, Fundamentals of Instructing, Technical Subject Areas, Preflight Preparation, and Emergency Operations. ACS Task code FI.I.B.K3, AI.II.D.K2, AI.II.E.K1c, AI.II.F.K3, AI.II.H.K1, AI.II.I.K3, AI.II.I.K4, AI.III.B.K1b, AI.XII.D.K1.

[Name], CFI Applicant

Ground lesson. Uses the matching slide deck "Missed Codes Review - Slides" (62 slides).

1. Objective

The student (or applicant) will explain each of the nine listed knowledge elements using FAA wording, apply each one to the Cessna 172S NAV III, and teach it back clearly.

ACS objective: To determine the applicant understands each of the listed knowledge elements and can explain them as an instructor.

2. Elements

ACS code	Element	Slides
FI.I.B.K3	Perceptions and insight.	4, 5, 6, 7, 8, 9
AI.II.D.K2	Airplane stability, maneuverability and controllability.	10, 11, 12, 13, 14, 15, 16
AI.II.E.K1c	Aircraft flight controls and operation of systems: powerplant and propeller.	17, 18, 19, 20, 21, 22, 60
AI.II.F.K3	Weight and balance terms, including: basic empty weight, maximum gross weight, arm, moment, reference datum, center of gravity (CG) and CG limits, and useful load.	23, 24, 25, 26, 27, 28, 29, 60
AI.II.H.K1	Ground-based navigation (identification, orientation, course determination, equipment, tests, regulations, interference, appropriate use of navigation data, and signal integrity).	30, 31, 32, 33, 34, 35, 36, 37, 60
AI.II.I.K3	Plotting a course.	38, 39, 40, 41, 42, 43
AI.II.I.K4	Power setting selection.	44, 45, 46, 47, 48, 60
AI.III.B.K1b	General airworthiness requirements: required inspections and aircraft logbook documentation.	49, 50, 51, 52, 53
AI.XII.D.K1	Emergency Locator Transmitter (ELT) operations, limitations, and testing requirements.	54, 55, 56, 57, 58, 59

3. Schedule

Part	Slides	Time
Introduction	Slides 1 to 3	5 min
Perceptions and insight	Slides 4 to 7	8 min
Stability and control	Slides 8 to 12	8 min
Powerplant and propeller	Slides 13 to 16	8 min
Weight and balance terms	Slides 17 to 21	8 min
Ground-based navigation	Slides 22 to 27	10 min
Plotting and power settings	Slides 28 to 34	10 min
Inspections and logbooks	Slides 35 to 37	6 min
ELT	Slides 38 to 41	6 min
Fixes, review, references	Slides 42 to 44	6 min

Total		75 min

4. Equipment

- Slide deck (44 slides)
- Whiteboard and markers
- C172S POH/AFM and aircraft logbooks
- Sectional chart, plotter, E6B
- AIM and 14 CFR Part 91
- Model airplane

5. Introduction

Attention: Ask: "Your student asks why the airplane drifts back to level by itself, why the VOR needle went backward, and whether the ELT is legal. Can you answer all three right now?"

Motivation: These nine codes were missed. Each one comes up again on the oral and when teaching real students. Knowing them in FAA wording makes the checkride smoother.

Overview: We will go through each code in ACS order: definition, plain words, C172S example. Then we fix the places where the study guide differs from the FAA books, and finish with review questions.

6. Development

Teach each slide in order. Slide numbers match the deck exactly.

Slide 1: Missed ACS Codes Review

ACS: Review

Use this slide for the attention step. Ask the student which of the nine topics they think is hardest, and start a quick list on the whiteboard.

Source: PHAK Fig 16-29

Slide 2: Objective

ACS: Review

State the objective: explain each of the nine knowledge elements using FAA wording, apply it to the C172S, and teach it back.

Source: ACS FI.I.B, ACS AI.II, ACS AI.III.B, ACS AI.XII.D

Slide 3: The Nine Codes

ACS: Review

Walk through the list. Each code gets two to five slides: the FAA definition, plain words, and a C172S example.

Source: FAA-S-ACS-25

Slide 4: Perceptions and Insight

ACS: FI.I.B.K3

Section start: Perceptions and Insight. Goals: What perceptions are; Five factors that affect perception; What insight is; How you build insight.

Source: AIH 3-7 to 3-9

Slide 5: Definition: Perceptions

ACS: FI.I.B.K3

Read the AIH 3-7 wording. Figure 3-5 shows most learning occurs through sight, and the combination of sight and hearing accounts for about 88 percent of all perception (AIH Fig 3-5).

Perception also involves the person giving meaning to what they sense. A new learner can be overwhelmed and focus on meaningless things, so the instructor directs the learner's attention to what matters (AIH 3-7).

Example: Teaching the G1000 engine page: point at the tach, say the number out loud, and have the student touch the knob. That uses sight, hearing, and touch together.

Source: AIH 3-7, AIH Fig 3-5

Slide 6: Factors That Affect Perception

ACS: FI.I.B.K3

AIH 3-7 lists five factors: physical organism, goals and values, self-concept, time and opportunity, and element of threat. The study guide memory aid GSTEP covers the same five in a different order (GSTEP is not an AIH term).

Physical organism: the senses a pilot needs to see, hear, feel, and respond (AIH 3-8).

Goals and values: perceptions depend on what the learner values (AIH 3-8).

Self-concept: a positive self-image keeps the learner open to new experiences. A negative one creates barriers (AIH 3-8).

Time and opportunity: it takes time and the right order of experiences to perceive (AIH 3-8).

Element of threat: fear narrows the perceptual field, so threats do not help learning (AIH 3-8).

Example: AIH example: during the first steep turns, a learner may stare at the altimeter and ignore outside references. That is the element of threat narrowing perception (AIH 3-8).

Ask the student:

- A learner just had a rough landing and says "I am bad at this." Which factor is at work?

Source: AIH 3-7, AIH 3-8

Slide 7: Definition: Insight

ACS: FI.I.B.K3

Read the AIH 3-9 wording. Insight is when the student sees how separate pieces relate to each other and to the whole task.

Example: AIH example made for the C172S: with a fixed-pitch propeller, rpm goes up when the throttle is opened and down when it is closed. Rpm also changes when pitch attitude changes with no power change. Engine speed, power, airspeed, and attitude are all related (AIH 3-9).

Source: AIH 3-9, PHAK Fig 7-8

Slide 8: How Instructors Build Insight

ACS: FI.I.B.K3

Key steps from AIH 3-9: point out relationships as they occur, provide a secure and nonthreatening environment, and help the learner keep a favorable self-concept.

Insight almost always happens eventually, but instruction speeds it up by teaching how perceptions relate as they occur (AIH 3-9).

Ask the student:

- In a C172S, you pitch up with no throttle change. What happens to rpm and airspeed? How does that build insight?

Source: AIH 3-9

Slide 9: Quick Check

ACS: FI.I.B.K3

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
Which sense accounts for most learning?	Sight, about 75 percent (AIH Fig 3-5).
Your student stares at the altimeter during their first steep turns. Which factor is at work?	Element of threat. Fear narrows perception (AIH 3-8).
What is insight?	Grouping perceptions into meaningful wholes (AIH 3-9).

Source: AIH 3-7 to 3-9, AIH Fig 3-5

Slide 10: Stability, Maneuverability, and Controllability

ACS: AI.II.D.K2

Section start: Stability, Maneuverability, and Controllability. Goals: Static vs. dynamic stability; Positive, neutral, negative; Maneuverability and controllability; How CG changes all three.

Source: PHAK 5-14, PHAK 5-15

Slide 11: Definition: Stability

ACS: AI.II.D.K2

Read the PHAK 5-14 definition word for word. The PHAK says the two types of stability are static and dynamic.

Source: PHAK 5-14, PHAK Fig 5-21

Slide 12: Static Stability

ACS: AI.II.D.K2

Static stability is the initial tendency, or direction of movement, back to equilibrium (PHAK 5-14).

Positive static stability: the initial tendency to return to the original state of equilibrium. Neutral: the tendency to remain in a new condition. Negative: the tendency to continue away from the original state (PHAK 5-14).

Analogy: the PHAK figure shows a ball in a bowl (positive), on a flat surface (neutral), and on top of a hill (negative).

Source: PHAK 5-14, PHAK Fig 5-21

Slide 13: Dynamic Stability

ACS: AI.II.D.K2

Dynamic stability refers to the aircraft response over time when disturbed (PHAK 5-14).

Positive: the motion decreases in amplitude and returns toward equilibrium. Neutral: the motion neither decreases nor increases. Negative: the motion increases and becomes more divergent (PHAK 5-15).

Analogy: Analogy from the PHAK: a worn automobile shock absorber shows neutral dynamic stability. The car keeps bouncing the same amount (PHAK 5-15).

Ask the student:

- Can an airplane have positive static stability but negative dynamic stability? (Yes. It starts back toward level, then overshoots more each time.)

Source: PHAK 5-14, PHAK 5-15, PHAK Fig 5-22

Slide 14: Definitions: Maneuverability and Controllability

ACS: AI.II.D.K2

Read both definitions from PHAK 5-15 word for word. The PHAK says stability in an aircraft affects these two areas significantly.

Source: PHAK 5-15, PHAK Fig 5-63

Slide 15: Putting the Three Together

ACS: AI.II.D.K2

A trainer like the C172S is quick to respond to control inputs, while a transport feels heavy and responds slowly. Designers balance these qualities for the airplane's purpose (PHAK 5-14).

CG location ties it together: an aircraft becomes less controllable at slow speeds and less stable as the CG moves aft (PHAK 5-43, PHAK 5-44).

Ask the student:

- Why does a trainer need more stability than an aerobatic airplane?

Source: PHAK 5-15, PHAK 5-43, PHAK 5-44

Slide 16: Quick Check

ACS: AI.II.D.K2

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
A ball resting on a flat table shows which static stability?	Neutral. It stays in its new position (PHAK 5-14).
The airplane oscillates and each swing gets bigger. What is that?	Negative dynamic stability (PHAK 5-15).
The CG moves aft. More stable or less stable?	Less stable (PHAK 5-44).

Source: PHAK 5-14, PHAK 5-15, PHAK 5-44

Slide 17: Powerplant and Propeller

ACS: AI.II.E.K1c

Section start: Powerplant and Propeller. Goals: The C172S engine; Throttle and mixture; How a propeller works; Fixed pitch and the tachometer.

Source: POH 7-28, POH 7-36, PHAK 7-1 to 7-5

Slide 18: C172S Engine

ACS: AI.II.E.K1c

An aircraft engine, or powerplant, produces thrust to propel an aircraft. Reciprocating engines work with a propeller to produce thrust (PHAK 7-1).

C172S: horizontally opposed, four cylinder, overhead valve, air-cooled, fuel-injected engine with a wet sump lubrication system. Lycoming IO-360-L2A rated at 180 horsepower at 2,700 RPM (POH 7-28, TCDS 3A12 p30).

Horizontally opposed engines have high power-to-weight ratios and a small frontal area (PHAK 7-2).

Source: POH 7-28, TCDS 3A12 p30, PHAK 7-2

Slide 19: Engine Controls

ACS: AI.II.E.K1c

Throttle: open full forward, closed full aft. In a fixed-pitch airplane, rpm is regulated by the throttle, which controls the fuel-air flow to the engine (POH 7-28, PHAK 7-5).

Mixture: red knob with a lock button. Rich is full forward, idle cutoff is full aft (POH 7-28).

Study guide fix: the guide says the throttle regulates fuel flow and lists a propeller control. The PHAK says fuel-air flow, and the C172S has a fixed-pitch propeller with no propeller control.

Source: POH 7-28, PHAK 7-5

Slide 20: Definitions: Propeller and Fixed Pitch

ACS: AI.II.E.K1c

Read both definitions. The propeller is twisted, with the highest pitch at the hub and the smallest at the tip, to make uniform lift along the blade because the tip travels faster than the hub (PHAK 7-4).

Thrust depends on the airfoil shape, the blade angle of attack, and engine rpm (PHAK 7-4).

Source: PHAK 7-4, PHAK 7-5, PHAK Fig 7-6

Slide 21: C172S Propeller

ACS: AI.II.E.K1c

C172S: two bladed, fixed pitch, one-piece forged aluminum alloy propeller, 76 inches in diameter (POH 7-36).

A fixed-pitch propeller is most efficient only at one combination of airspeed and rpm, so it is ideal for neither cruise nor climb. It is used when low weight, simplicity, and low cost are needed (PHAK 7-5).

Propeller efficiency is the ratio of thrust horsepower to brake horsepower (PHAK 5-29). In plain words: the engine makes power, and the propeller turns part of it into thrust.

Source: POH 7-36, PHAK 7-5, PHAK 5-29

Slide 22: Quick Check

ACS: AI.II.E.K1c

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
What engine does the C172S have, and how much power?	Lycoming IO-360-L2A, 180 hp at 2,700 RPM (POH 7-28).
What instrument shows engine power in the C172S?	The tachometer (PHAK 7-5).
Why is the propeller blade twisted?	For even lift from hub to tip, because the tip travels faster (PHAK 7-4).

Source: POH 7-28, PHAK 7-4, PHAK 7-5

Slide 23: Weight and Balance Terms

ACS: AI.II.F.K3

Section start: Weight and Balance Terms. Goals: Datum and arm; Moment and CG; Basic empty weight and useful load; CG limits and maximum weight.

Source: PHAK 10-4, PHAK 10-5

Slide 24: Definitions: Datum and Arm

ACS: AI.II.F.K3

Read the PHAK definitions word for word. C172S datum: lower portion of the front face of the firewall (TCDS 3A12 p30).

Study guide fix: the guide gives shortened definitions. Use the PHAK wording on the checkride.

Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-2

Slide 25: Definitions: Moment and CG

ACS: AI.II.F.K3

Read the definitions. Example: 50 lb at 100 inches = 5,000 in-lb (PHAK 10-6). To find CG, divide total moment by total weight (PHAK 10-7).

Source: PHAK 10-4, PHAK 10-5, PHAK Fig 10-3

Slide 26: Definitions: Basic Empty Weight and Useful Load

ACS: AI.II.F.K3

Read the definitions. POH sample: basic empty weight 1,642 lb, which includes unusable fuel and full oil (POH 6-10). 2,550 minus 1,642 = 908 lb useful load.

Source: PHAK 10-4, PHAK 10-5, POH Fig 6-3

Slide 27: Definitions: CG Limits and Maximum Weight

ACS: AI.II.F.K3

Read the definitions. The ACS says "maximum gross weight"; the PHAK term is "maximum weight."

Source: PHAK 10-5, POH 2-8

Slide 28: C172S Weight and Balance Numbers

ACS: AI.II.F.K3

Normal category: maximum takeoff and landing weight 2,550 lb, ramp 2,558 lb (POH 2-8). Forward CG limit 35.0 inches at 1,950 lb or less, straight line to 41.0 inches at 2,550 lb. Aft limit 47.3 inches at all weights (POH 2-8, TCDS 3A12 p30).

Full detail and worked problems are in the II.F Performance and Limitations lesson.

Source: TCDS 3A12 p30, POH 2-8, PHAK 10-7

Slide 29: Quick Check

ACS: AI.II.F.K3

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
Where is the C172S datum?	Lower portion of the front face of the firewall (TCDS 3A12 p30).
A 120 lb bag is 95 inches aft of the datum. What is its moment?	120 x 95 = 11,400 in-lb.
Basic empty weight is 1,642 lb and maximum weight is 2,550 lb. What is the useful load?	908 lb.

Source: TCDS 3A12 p30, PHAK 10-6, POH 6-10

Slide 30: Ground-Based Navigation

ACS: AI.II.H.K1

Section start: Ground-Based Navigation. Goals: Identify a VOR; Read the CDI and HSI; Track a radial in wind; VOR checks, NDB, interference.

Source: AIM 1-1-2 to 1-1-4, PHAK 16-23 to 16-26

Slide 31: VOR: Identification and Accuracy

ACS: AI.II.H.K1

VORs operate from 108.0 to 117.95 MHz and are subject to line-of-sight limits. Range increases with altitude (AIM 1-1-3).

Identification: the only positive method is the Morse code identifier or the recorded voice identification with the word "VOR" after the name. During maintenance the station may send T-E-S-T or remove the identifier (AIM 1-1-3).

Accuracy: course alignment is generally plus or minus 1 degree (AIM 1-1-3).

Source: AIM 1-1-3, PHAK Fig 16-29

Slide 32: Orientation: CDI and HSI

ACS: AI.II.H.K1

The CDI has an omnibearing selector (OBS), a course deviation needle, and a TO/FROM indicator (PHAK 16-23).

The HSI combines the heading indicator and the CDI in one display, which is how the G1000 PFD presents it (PHAK 16-24).

Reverse sensing: fly the selected course with a TO indication when going to a station and FROM when going away. If not, the needle works backward (PHAK 16-26).

Study guide fix: the guide says to correct reverse sensing by turning away from the needle. The PHAK way is to set the course so the TO/FROM matches your direction, and then the needle reads normally.

Source: PHAK 16-23, PHAK 16-24, PHAK 16-26, PHAK Fig 16-30

Slide 33: Course Determination: Tracking

ACS: AI.II.H.K1

PHAK example: holding 350 degrees with wind from the right, the airplane drifts left and the needle moves right. Turn right to return, then turn back toward the course with a wind correction (PHAK 16-25).

Just turning toward the needle causes overshooting and S-turns. Fly headings that re-intercept, then apply a drift correction. If the needle fluctuates slightly, wait before changing heading (PHAK 16-26).

Station passage: the needle fluctuates, then the TO changes to FROM (PHAK 16-25).

Source: PHAK 16-25, PHAK 16-26, PHAK Fig 16-32

Slide 34: Tests and Regulations

ACS: AI.II.H.K1

14 CFR 91.171 requires a VOR check within the preceding 30 days for IFR. Limits: VOT or ground checkpoint plus or minus 4 degrees, airborne checkpoint plus or minus 6 degrees, dual VOR check 4 degrees between the two (14 CFR 91.171).

Using a VOT: with the CDI centered, the OBS should read 0 degrees FROM or 180 degrees TO (AIM 1-1-4).

Locations of VOTs and checkpoints are in the Chart Supplement (AIM 1-1-4).

Source: 14 CFR 91.171, AIM 1-1-4

Slide 35: NDB and ADF

ACS: AI.II.H.K1

NDBs normally operate from 190 to 535 kHz and send a continuous three-letter Morse identifier, except compass locators (AIM 1-1-2).

ADF receivers do not have a flag to warn of bad bearing information, so the pilot should continuously monitor the NDB identifier (AIM 1-1-2).

The RMI shows ADF and VOR bearings with pointers on a compass card (PHAK 16-25).

Source: AIM 1-1-2, PHAK 16-25, PHAK Fig 16-31

Slide 36: Interference and Signal Integrity

ACS: AI.II.H.K1

Certain propeller rpm settings can make the CDI fluctuate as much as plus or minus 6 degrees. Slight rpm changes normally smooth it out. Check this before reporting a station or receiver problem (AIM 1-1-3).

Minor course roughness or brief flag activity can occur, especially in mountains. Use the TO/FROM indicator to confirm station passage (AIM 1-1-3).

NDB bearings can be disturbed by lightning, precipitation static, and distant stations at night (AIM 1-1-2).

Appropriate use of navigation data: when equipment decodes and displays the identifier, compare it to the chart (AIM 1-1-3).

Source: AIM 1-1-2, AIM 1-1-3

Slide 37: Quick Check

ACS: AI.II.H.K1

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
What is the only positive way to identify a VOR?	Its Morse code identifier or recorded voice ID with the word "VOR" (AIM 1-1-3).
With the CDI centered on a VOT, what should the OBS read?	0 FROM or 180 TO (AIM 1-1-4).
The CDI swings back and forth at one RPM. What do you try first?	Change the RPM slightly (AIM 1-1-3).
Why must you keep listening to an NDB identifier?	The ADF has no warning flag for bad bearings (AIM 1-1-2).

Source: AIM 1-1-3, AIM 1-1-4, AIM 1-1-2

Slide 38: Plotting a Course

ACS: AI.II.I.K3

Section start: Plotting a Course. Goals: True course; Using the plotter; Magnetic variation; True to magnetic.

Source: PHAK 16-5 to 16-7, PHAK 16-12, PHAK 16-18

Slide 39: Definition: True Course

ACS: AI.II.I.K3

Read the definition. Direction is measured clockwise from true north using the meridians (PHAK 16-5). In the figure, A to B is 065 degrees and the return trip is 245 degrees.

Source: PHAK 16-5, PHAK Fig 16-7

Slide 40: Plotting the Course

ACS: AI.II.I.K3

Draw the course line from the center of the departure airport to the center of the destination (PHAK 16-18).

Because meridians converge toward the poles, measure the course at a meridian near the midpoint of the course (PHAK 16-5).

A plotter is a protractor and ruler. Most have NM and SM scales, with a sectional scale on one side and a world aeronautical chart scale on the other (PHAK 16-12). Use the sectional scale on a sectional chart.

Use nautical miles so distance matches knots for time and fuel. The PHAK fuel example: 400 NM at 100 knots GS = 4 hours, at 5 GPH = 20 gallons (PHAK 16-12).

Ask the student:

- Your airspeed is in knots. Which plotter scale do you use for distance?

Source: PHAK 16-5, PHAK 16-12, PHAK 16-18, PHAK Fig 16-25

Slide 41: Definition: Variation

ACS: AI.II.I.K3

Read the definition. Variation is shown on charts as broken magenta isogonic lines connecting points of equal variation. The line of zero variation is the agonic line (PHAK 16-6).

On the west coast the compass points east of true north; on the east coast it points west of true north (PHAK 16-6).

Source: PHAK 16-6, PHAK Fig 16-9

Slide 42: True Course to Magnetic Course

ACS: AI.II.I.K3

Convert TC to magnetic course by adding or subtracting the variation from the nearest isogonic line. If variation is west, add. If east, subtract (PHAK 16-7).

PHAK example: variation 9 degrees E. True north 360 becomes magnetic 351. True east 090 becomes magnetic 081 (PHAK 16-7).

If you cross isogonic lines, use the new value from that point on (PHAK 16-7).

Ask the student:

- TC is 220 and variation is 6 degrees W. What is the magnetic course? (226)

Source: PHAK 16-7

Slide 43: Quick Check

ACS: AI.II.I.K3

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
When you measure with a plotter, are you referencing true or magnetic north?	True north. The result is the true course (PHAK 16-5).
True course 090, variation 9 degrees east. Magnetic course?	081 (PHAK 16-7).
On a long route, where do you measure the course?	At a meridian near the midpoint (PHAK 16-5).

Source: PHAK 16-5, PHAK 16-7

Slide 44: Power Setting Selection

ACS: AI.II.I.K4

Section start: Power Setting Selection. Goals: Use the POH cruise tables; Power in a fixed-pitch airplane; Read the chart notes; Speed versus fuel.

Source: POH 4-33 to 4-35, POH Fig 5-8, PHAK 7-5

Slide 45: Choosing a Power Setting

ACS: AI.II.I.K4

The Cruise Performance Table shows true airspeed and nautical miles per gallon at various altitudes and percent powers, for standard conditions and zero wind. Use it with winds aloft to find the most favorable altitude and power setting (POH 4-33).

The POH says choosing altitude for the best winds and using low power settings are significant factors to reduce fuel use on every trip (POH 4-33).

Example from Figure 4-3: at 8,000 feet, 75 percent gives 124 KTAS at 12.2 NMPG and 55 percent gives 107 KTAS at 13.6 NMPG (POH 4-33).

Fuel burn also depends on engine condition, rpm, mixture, and percent of horsepower used (PHAK 16-12).

Source: POH 4-33, PHAK 16-12

Slide 46: Power in a Fixed-Pitch Airplane

ACS: AI.II.I.K4

In a fixed-pitch propeller airplane, the tachometer is the indicator of engine power (PHAK 7-5).

Power depends on air density. 2,300 rpm at 5,000 feet produces less horsepower than 2,300 rpm at sea level (PHAK 7-5).

The POH cruise data is based on the recommended lean mixture at 75 percent Maximum Continuous Power (MCP) and less (POH 4-34). Use full rich mixture above 75 percent power (POH 4-35).

Study guide fix: the guide explains brake horsepower as power measured with a brake. That explanation is not in the FAA handbooks. The PHAK uses brake horsepower as engine power and says propeller efficiency is thrust horsepower divided by brake horsepower (PHAK 5-29).

Source: PHAK 7-5, POH 4-34, POH 4-35

Slide 47: Example: Read the Notes

ACS: AI.II.I.K4

POH sample: at 8,000 feet and 20 degrees C above standard, 2,600 RPM gives 64 percent power, 117 KTAS, and 8.9 GPH (POH 5-5).

Figure 5-8 note: maximum cruise power using recommended lean mixture is 75 percent MCP. Values above 75 percent MCP are shown for interpolation only. Operations above 75 percent MCP must use full rich mixture (POH Fig 5-8).

Teach the learner to read the whole chart, including conditions and notes, before choosing a setting.

Ask the student:

- You want 2,700 RPM at 2,000 feet. What does the note tell you about mixture?

Source: POH 5-5, POH Fig 5-8

Slide 48: Quick Check

ACS: AI.II.I.K4

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
What should your student look at first to pick a power setting?	The POH cruise performance table and charts (POH 4-33).
Above 75 percent power, how do you set the mixture?	Full rich (POH Fig 5-8).
Same RPM, higher altitude: more or less power?	Less, because the air is less dense (PHAK 7-5).

Source: POH 4-33, POH Fig 5-8, PHAK 7-5

Slide 49: Inspections and Logbooks

ACS: AI.III.B.K1b

Section start: Inspections and Logbooks. Goals: Annual and 100-hour; Transponder, ELT, other checks; ADs; What goes in the logbook.

Source: 14 CFR 91.409, 91.413, 91.417

Slide 50: Required Inspections

ACS: AI.III.B.K1b

Annual: no person may operate an aircraft unless it has had an annual inspection within the preceding 12 calendar months (14 CFR 91.409). Twelve calendar months runs from any day of a month to the last day of the same month the next year (PHAK 9-8).

100-hour: required to carry any person (other than crew) for hire, or to give flight instruction for hire in an aircraft that person provides. It may be exceeded by not more than 10 hours to reach a place where it can be done, and the extra time counts toward the next 100 hours (14 CFR 91.409).

The owner or operator is primarily responsible for keeping the aircraft airworthy, including compliance with ADs under part 39 (14 CFR 91.403). The POH says the FAA may require other inspections through ADs (POH 8-7).

Note: an amendment to 91.403 and 91.409 was published November 10, 2026 and is not yet in the current text. Recheck before your checkride.

Source: 14 CFR 91.409, 14 CFR 91.403, PHAK 9-8, POH 8-7

Slide 51: Other Required Checks

ACS: AI.III.B.K1b

Transponder: tested and inspected within the preceding 24 calendar months (14 CFR 91.413).

ELT: inspected within 12 calendar months after the last inspection (14 CFR 91.207).

Altimeter and static system: tested within 24 months to operate in controlled airspace under IFR (PHAK 9-9).

VOR: checked within the preceding 30 days for IFR (14 CFR 91.171).

Source: 14 CFR 91.413, 14 CFR 91.207, 14 CFR 91.171, PHAK 9-9

Slide 52: Logbook Documentation

ACS: AI.III.B.K1b

Maintenance records must include a description of the work, the date completed, and the signature and certificate number of the person approving return to service (14 CFR 91.417).

Records must also show total time in service, status of life-limited parts, time since overhaul, current inspection status, and current AD status with method of compliance (14 CFR 91.417).

Work records are kept until the work is repeated or superseded, or for 1 year. Total time, AD status, and similar records are kept and transferred with the aircraft when sold (14 CFR 91.417).

C172S POH: the airplane logbook and engine logbook are to be made available on request (POH 8-7).

Source: 14 CFR 91.417, POH 8-7

Slide 53: Quick Check

ACS: AI.III.B.K1b

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
How often is an annual inspection required?	Within the preceding 12 calendar months (14 CFR 91.409).
You give paid instruction in an airplane you provide. What other inspection applies?	The 100-hour inspection (14 CFR 91.409).
How often must the transponder be checked?	Every 24 calendar months (14 CFR 91.413).
What three things must a maintenance entry include?	Description of work, date, and the signature and certificate number of the person approving return to

	service (14 CFR 91.417).

Source: 14 CFR 91.409, 91.413, 91.417

Slide 54: Emergency Locator Transmitter

ACS: AI.XII.D.K1

Section start: Emergency Locator Transmitter. Goals: How the ELT works; Using the C172S ELT; Limitations; Testing and inspection rules.

Source: AIM 6-2-4, 14 CFR 91.207, POH Supplement 1

Slide 55: ELT Operations

ACS: AI.XII.D.K1

ELTs operate on 121.5 MHz, 243.0 MHz, or 406 MHz. The 406 MHz ELT is digital and can carry owner and aircraft data (AIM 6-2-4).

When armed, ELTs activate automatically from crash forces and transmit continuously for at least 48 hours (AIM 6-2-4).

406 MHz ELTs must be registered with NOAA (AIM 6-2-4).

C172S (this POH): Pointer Model 4000-11. AUTO arms the G switch, ON activates it for test or emergency, OFF/RESET deactivates it (POH S1-7).

Source: AIM 6-2-4, POH S1-7

Slide 56: Using the C172S ELT

ACS: AI.XII.D.K1

The remote switch and annunciator are on the top center of the right front passenger panel. The light comes on when the ELT transmits (POH 7-71).

Before a forced landing, especially in remote or mountain areas, set the remote switch to ON (POH S1-8).

After landing: verify activation and listen on 121.5. Save the battery until you see rescue aircraft. Then set RESET and release to AUTO, and try to call on 121.5. If no contact, go back to ON (POH S1-9).

Source: POH 7-71, POH S1-8, POH S1-9

Slide 57: ELT Limitations

ACS: AI.XII.D.K1

Cospas-Sarsat stopped monitoring 121.5 and 243.0 MHz in 2009. Those ELTs depend on ATC or an overflying aircraft hearing the signal (AIM 6-2-4).

SAR forces wait for confirmation on 121.5/243.0 alerts, which can take hours, but can respond to 406 MHz alerts in minutes (AIM 6-2-4).

An ELT helps only if it survives the crash and activates (AIM 6-2-4).

False alarms cause expensive searches and can hide real signals. Monitor 121.5 before engine shutdown at the end of each flight (AIM 6-2-4).

Source: AIM 6-2-4

Slide 58: ELT Testing and Rules

ACS: AI.XII.D.K1

Analog 121.5/243 ELTs: test only during the first 5 minutes after any hour, for no more than three audible sweeps. Digital 406 MHz ELTs: test per the manufacturer. Airborne tests are not authorized (AIM 6-2-4).

C172S POH: test every 100 hours, within the first 5 minutes of the hour, permitting only three emergency tones (POH S1-10).

14 CFR 91.207: inspect within 12 calendar months for installation, battery corrosion, controls and crash sensor, and signal. Replace or recharge the battery after more than 1 cumulative hour of use or at 50 percent of its useful life. Mark the expiration date on the transmitter and in the maintenance record.

Exceptions include training operations conducted entirely within 50 nautical miles of the airport where they began, and an ELT removed for repair for up to 90 days with a logbook entry and an "ELT not installed" placard (14 CFR 91.207).

Source: AIM 6-2-4, 14 CFR 91.207, POH S1-10

Slide 59: Quick Check

ACS: AI.XII.D.K1

Quick check. Ask each question, wait for the answer, then confirm.

Question	Answer
When may you test an analog ELT?	In the first 5 minutes after the hour, no more than three sweeps (AIM 6-2-4).
When must the ELT battery be replaced?	After more than 1 cumulative hour of use, or at 50 percent of its useful life (14 CFR 91.207).
You are about to make a forced landing in the mountains. What do you do with the ELT?	Set the remote switch to ON (POH S1-8).

Source: AIM 6-2-4, 14 CFR 91.207, POH S1-8

Slide 60: Study Guide Fixes

ACS: AI.II.E.K1c, AI.II.F.K3, AI.II.H.K1, AI.II.I.K4

Throttle controls fuel-air flow (PHAK 7-5).

The C172S has a fixed-pitch propeller and no propeller control (POH 7-36).

For reverse sensing, set the OBS so TO/FROM matches your direction of flight (PHAK 16-26).

Use PHAK definitions for weight and balance terms (PHAK 10-4, PHAK 10-5).

Brake horsepower: teach it as engine power. The "measured with a brake" explanation is not in the FAA handbooks (PHAK 5-29).

Source: PHAK 7-5, POH 7-36, PHAK 16-26, PHAK 10-4, PHAK 5-29

Slide 61: Review Questions

ACS: Review

Answers: (1) physical organism, goals and values, self-concept, time and opportunity, element of threat (AIH 3-7). (2) Positive returns to equilibrium; neutral stays in the new condition (PHAK 5-14). (3) Subtract (PHAK 16-7). (4) First 5 minutes after the hour, no more than three sweeps (AIM 6-2-4).

Extra questions: What indicates engine power in a C172S? (Tachometer, PHAK 7-5.) Who must keep the maintenance records? (Owner or operator, 14 CFR 91.417.) Which plotter scale do you use with knots? (Nautical miles.)

Source: AIH 3-7, PHAK 5-14, PHAK 16-7, AIM 6-2-4

Slide 62: References

ACS: Review

Review the references. The study guide was the starting outline, and every answer was checked against these FAA sources.

Source: See lesson plan for full list

7. Instructor Actions

- Teach each code with the definition first, then plain words, then the C172S example.
- Use the whiteboard for the stability ball analogy, the VOR tracking picture, and the variation math.
- Show the actual POH and logbooks if available.
- Ask the questions on each slide and have the student teach one code back.

8. Student Actions

- Follow along and take notes.
- Answer the questions on each slide.
- Teach back at least two codes in their own words.
- Find the inspection entries in a real logbook.

9. Common Errors

Using paraphrased definitions. Use the FAA wording from the PHAK and AIH.

Saying the throttle controls fuel flow. It controls fuel-air flow (PHAK 7-5).

Describing a propeller control for the C172S. The C172S has a fixed-pitch propeller (POH 7-36).

Adding east variation. East is least: subtract. West is best: add (PHAK 16-7).

Measuring distance in statute miles with knots. Use the nautical mile scale (PHAK 16-12).

Testing an ELT any time. First 5 minutes after the hour, three sweeps max (AIM 6-2-4).

Forgetting the 100-hour rule for instruction. 100-hour applies to flight instruction for hire in an aircraft the instructor provides (14 CFR 91.409).

10. Conclusion and Review

Summarize each code in one sentence: perceptions come through the senses and insight groups them; stability returns the airplane to equilibrium; the C172S has a 180 hp Lycoming and a fixed-pitch propeller; weight and balance terms come from the PHAK; VOR use depends on proper ID and TO/FROM; plot true course and convert with variation; pick power from the POH and read the notes; annual, 100-hour, ADs, and records; ELT test and inspection rules.

Use slide 43 for review, then have the student teach one code back with no slides.

11. Completion Standards

- Explains each of the nine elements using FAA wording.
- Applies each element to the C172S where it fits.
- Answers the review questions without help.
- Teaches at least two codes back clearly.

12. References

- FAA-S-ACS-25, Flight Instructor for Airplane Category ACS (November 2023): the nine element codes listed above.
- FAA-H-8083-9, Aviation Instructor's Handbook (AIH): 3-7 to 3-9.
- FAA-H-8083-25C, Pilot's Handbook of Aeronautical Knowledge (PHAK): 5-14, 5-15, 5-29, 5-43, 5-44, 7-1, 7-2, 7-4, 7-5, 9-8, 9-9, 10-4 to 10-7, 16-5 to 16-7, 16-12, 16-18, 16-23 to 16-26. The October 2025 addendum does not change these pages.
- Aeronautical Information Manual (AIM), Basic with Changes 1, 2, and 3 (Change 3 effective July 9, 2026): 1-1-2, 1-1-3, 1-1-4, 6-2-4.
- 14 CFR 91.171, 91.207, 91.403, 91.409, 91.413, 91.417 (eCFR, current as of November 10, 2026). Note: an amendment to 91.403 and 91.409 was published November 10, 2026 (91 FR 72405) and is not yet in the current text. It appears to cover inspection programs for single-engine turbine and unmanned aircraft. Recheck the text before your checkride.
- Cessna 172S NAV III (G1000) POH pages, P/N 172SPHAUS, from Information Manual 172SIMAUS Rev 3: 2-8, 4-33 to 4-35, 5-19 (Figure 5-8), 6-10, 7-28, 7-36, 7-71, 8-7, and Section 9 Supplement 1 (S1-7 to S1-10). Training use only.
- FAA Type Certificate Data Sheet 3A12, Revision 89: page 30.
- Starting outline: ATP CFI Study Guide pages supplied by the applicant. Answers were checked against the FAA sources above and corrected where they differ.